

OptiscanAI: Large Vision Models for Multi-Disease Retinal Screening in Ugandan Healthcare

A Benchmark and Deployment Study

Lauben Mpairwe*, Shadia Nankya*, Rebecca Yapyeko*, Marvin Ggaliwango†

* Department of Networks, College of Computing and Information Sciences
Makerere University, Kampala, Uganda

Email: {mpairwe_lauben, nankya.shadia, yapyeko.rebecca}@students.mak.ac.ug

† Department of Computer Science, College of Computing and Information Sciences
Makerere University, Kampala, Uganda

Email: ggaliwango.marvin@cit.ac.ug

Abstract—Preventable blindness from vitreoretinal and optic nerve pathologies presents a critical global burden, exacerbated by the acute shortage of specialist ophthalmologists across sub-Saharan Africa. This investigation delivers a rigorous empirical assessment of OptiscanAI, an integrated multi-label retinal screening framework adapted from a domain-specific foundation model, evaluated strictly on the held-out official test partition (640 images) of the Retinal Fundus Multi-Disease Image Dataset. We detail class-stratified sensitivity, specificity, positive predictive value (PPV), false-negative rate, and area under the precision-recall curve (AUPRC) with 95% bootstrap confidence intervals, alongside controlled ablations across network backbones, parameter-efficient adaptation, and dynamic 8-bit quantization. Our findings illuminate three operational realities for low-resource translation. First, discriminative capacity is robust yet unevenly distributed: macro AUC reaches 0.872, ranging from 0.996 for myopia to 0.663 for optic disc pallor. Second, uncalibrated probability outputs exhibit severe over-confidence (expected calibration error 0.209), causing static decision thresholds to silence five diagnostic channels; isotonic recalibration mitigates this error to 0.003 without altering discriminative ranking. Third, while binary referral triage achieves an AUC of 0.952, projecting this operating point to primary-care clinical prevalence drops PPV from 0.93 to below 0.30. The design emphasizes a human-in-the-loop paradigm, traceability, and an explicit separation between benchmark performance and clinical readiness.

Index Terms—retinal screening, multi-label classification, model calibration, explainability, low-resource deployment, foundation models

I. INTRODUCTION

Preventable blindness represents an escalating healthcare crisis: the World Health Organization estimates that 2.2 billion individuals endure visual impairment,

over half treatable if diagnosed early [1]. Microvascular complications such as diabetic retinopathy affected over 103 million patients by 2020 and continue to expand in developing regions [2]. In Uganda, specialist ophthalmologists are concentrated in urban referral centres, leaving non-specialist primary healthcare workers as the initial gatekeepers for ocular screening. Under these constraints, insidious pathologies—such as diabetic retinopathy, hypertensive remodeling, and glaucomatous neuropathy—are frequently missed until irreversible vision loss occurs.

Although deep neural models achieve high diagnostic accuracy on isolated conditions [3], and retinal foundation backbones exhibit versatile transfer [8], practical deployment in resource-constrained clinics faces critical challenges: severe class skew, clinical explainability demands, unstable telecommunications, and population distribution shift [4], [5].

In response to peer review of our initial report, this manuscript addresses the core concerns raised. Reviewers correctly observed that evaluation on an international benchmark (RFMiD) cannot support unqualified clinical claims in Ugandan healthcare, and requested explicit disentanglement of component contributions. We retain the system architecture but reposition our claims: what follows is benchmark evidence and deployment-constraint analysis, stating the gap to clinical readiness quantitatively rather than as a caveat.

Contributions. (i) A per-disease evaluation on the held-out RFMiD test split reporting sensitivity, specificity, PPV, FNR, and AUPRC with 95% bootstrap confidence intervals. (ii) A unified ablation isolating retinal pretraining, LoRA adaptation, decision thresholding, and INT8 quantization under a shared training recipe. (iii) Diagnosis of a calibration failure that silences five

disease channels, with a rank-preserving correction. (iv) Quantitative faithfulness testing of saliency maps and generated text via insertion/deletion metrics. (v) A transparent evaluation framework that explicitly separates benchmark performance from clinical readiness, mapping triage metrics onto primary-care referral burdens.

II. RELATED WORK

A. Retinal Detection and Foundation Models

Deep learning for retinal fundus analysis was established by Gulshan *et al.* [3] for diabetic retinopathy, and extended to glaucoma and macular degeneration, predominantly on single-disease tasks. Frontline screening requires multi-label classification across heterogeneous taxonomies, where class imbalance, co-morbidity, and positive scarcity impede learning [6]. Vision Transformers [7] capture global spatial dependencies across fundus patches, and RETFound [8], pretrained on 1.6 million retinal images, surpasses ImageNet initialisation on ophthalmic benchmarks. Low-Rank Adaptation (LoRA) [9] updates parameter subspaces while freezing backbone weights, enabling adaptation under constrained compute budgets.

B. Imbalance, Calibration and Explainability

Asymmetric [15] and focal losses [14] address imbalance by down-weighting easy negatives, but can distort probability calibration in over-confident networks [16]; recalibration via Platt scaling restores usable probabilities while preserving ranking. Grad-CAM [10] and Integrated Gradients [12] are standard attribution baselines, evaluated via perturbation metrics [13]. Field studies confirm that workflow ergonomics, image quality, and clinician trust dominate real-world effectiveness [4].

III. SYSTEM ARCHITECTURE

Fig. 1 shows the pipeline; the numbers below refer to its labelled components.

A. Input Quality Gating (1)

To enforce diagnostic quality, a cascaded four-tier intake filter screens photographs prior to inference: structural verification enforces minimum resolution and physiological aspect ratios; non-parametric statistical metrics compute illumination uniformity, focus sharpness, field-of-view centration, and circular mask integrity; a lightweight MobileNetV3-Small classifier estimates photographic usability; and an empirical decision boundary enforces an intake criterion of $0.6 s_{\text{stat}} + 0.4 s_{\text{learned}} \geq 0.70$. Deficient captures prompt immediate operator recapture guidance rather than generating ungrounded risk predictions.

B. Backbone and Adaptation (2)–(3)

The backbone is RETFound ViT-L/16 [8] operating at 224×224 , giving 196 patch tokens and a 1024-dimensional representation that is projected to 512. LoRA adapters of rank $r=16$ with $\alpha=32$ are injected into the query-key-value projection of each of the 24 transformer blocks, so

$$W' = W_0 + \frac{\alpha}{r} BA, \quad (1)$$

where W_0 is frozen and only $B \in \mathbb{R}^{d \times r}$ and $A \in \mathbb{R}^{r \times k}$ are trained. This leaves 2.44 M trainable parameters out of 305.7 M, or 0.8%. A bottleneck head ($512 \rightarrow 128 \rightarrow 24$, dropout 0.5 and 0.3) produces the multi-label output. Within the network, a single sparse top- k attention layer ($k=32$) over the patch tokens provides spatial context pooling before classification; we note explicitly that this is patch-level self-attention, *not* a graph attention network over the clinical knowledge graph. Classes with fewer than 10 training positives are dropped, reducing the RFMiD taxonomy from 45 labels to the 24 that can be learned at all.

C. Loss and Decision-Threshold Policy (4)

Screening tolerates false alarms far better than misses, but an unusable alarm rate destroys the referral pathway. Training uses an asymmetric loss [15] that never down-weights positives and aggressively suppresses easy negatives, with $\gamma_{\text{pos}}=0$, $\gamma_{\text{neg}}=4$, probability clipping at 0.05 and label smoothing 0.05. After training, a per-class threshold τ_c is fitted on the validation split as the lowest threshold whose precision still satisfies a floor of 0.10; classes that cannot meet the floor at any threshold are assigned $\tau_c=0.95$, which in practice disables them. Section V shows that this policy interacts badly with the model's calibration, and reports the correction.

D. Clinical Knowledge Graph (5)

The knowledge graph is a curated symbolic layer, not a learned module, and it is applied *after* classification. For the 24 deployed classes it holds 24 disease nodes and 54 typed edges over six relation families: co-occurrence, category membership (vascular, degenerative, glaucomatous, infectious/immunologic, neuro-ophthalmic, tractional), severity grade on a three-point scale, systemic association with weights, age-related risk, and treatment consideration. Content was extracted by the authors from clinical practice guidelines and primary literature, with the source recorded per disease: the American Academy of Ophthalmology Preferred Practice Pattern for primary open-angle glaucoma (2022) for optic disc cupping,

Server path (full model)

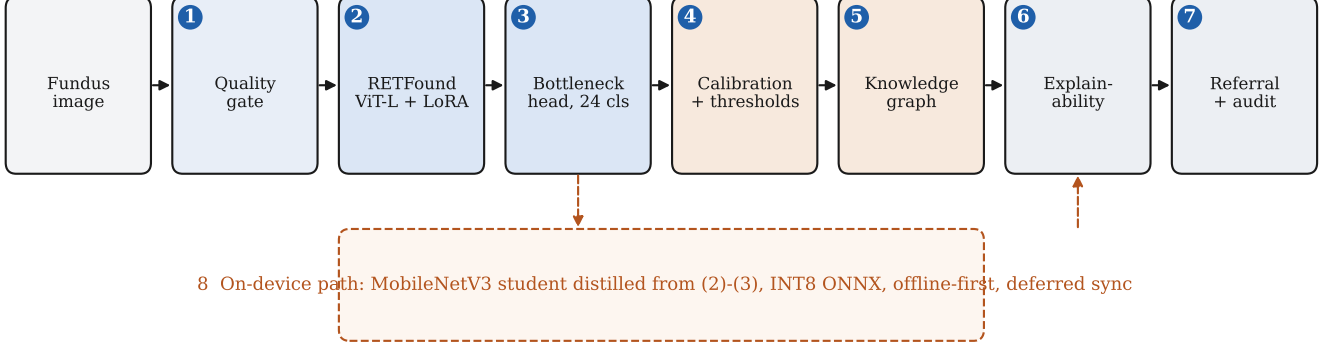


Fig. 1. OptiscanAI pipeline. (1) image quality gate; (2) RETFound ViT-L/16 backbone with LoRA adapters; (3) bottleneck classification head over 24 retained classes; (4) probability calibration and per-class decision thresholds; (5) clinical knowledge graph applied post-hoc; (6) explainability layer; (7) referral decision with an audit record. Component (8) is the offline path: a MobileNetV3 student distilled from (2)–(3), exported to INT8 ONNX, which runs on-device and defers synchronisation until connectivity returns.

AREDS Report No. 39 for age-related macular degeneration, and the Uganda Ministry of Health diabetes guidelines (2024) for diabetic retinopathy, with per-condition journal references for vein occlusion and retinal pigment epithelial changes carried in the released graph file. Uganda-specific prevalence weights are attached to six conditions. The graph has *not* been formally validated by an independent ophthalmologist panel; that review is outstanding and we do not claim it. Its function at inference is to refine co-occurring probabilities, assign a three-level referral priority and compute a composite risk score; Section V measures what that is worth.

E. Explainability (6) and Offline Deployment (8)

The interpretability suite combines Grad-CAM [10] visual saliency maps, Integrated Gradients [12] feature attributions, and concise clinical narratives synthesized by a distilled language model grounded in the disease ontology. For disconnected primary-care facilities, a compact MobileNetV3-Large student model (5.2 M parameters) is knowledge-distilled using a composite objective encompassing soft logits, hard ground-truth, intermediate feature alignment, and boundary-threshold consistency, and converted to an 8-bit INT8 ONNX binary (5.4 MB, compressed from 20.8 MB in FP32). Edge inference operates autonomously without telecommunications access, staging encrypted records for opportunistic synchronization upon network restoration. All clinical interactions are immutably logged for audit traceability.

IV. EXPERIMENTAL SETUP

Data. The empirical foundation rests on the Retinal Fundus Multi-Disease Image Dataset (RFMiD) [6], adhering strictly to its canonical three-way partition: 1,920 training, 640 validation, and 640 evaluation images, annotated across 45 clinical categories spanning severe prevalence skew (1% to 50%). Crucially, all reported performance figures reflect the 640-image held-out test split, evaluated strictly post-convergence; hyperparameter tuning, model selection, and operating thresholds were established exclusively on validation data.

Preprocessing. Images are resized directly to 224×224 with bicubic interpolation and normalised with ImageNet statistics. Two reviewer questions concern this choice, and rather than argue them we measured both as ablation arms in Table II. ImageNet statistics are retained because the backbone was pretrained under them: matching the *pretraining* input distribution matters more than matching the fine-tuning corpus, since the LoRA adapters modify a representation learned in that space. Substituting RFMiD’s own channel statistics costs 0.054 AUC and 0.107 AUPRC, which supports keeping them. We also confirm that the deployed pipeline applies *no* histogram equalisation; an earlier draft described one, and it is not in the implementation. Since the reviewer’s concern was that equalisation may distort colour-dependent findings, we tested CLAHE on the L channel of LAB rather than assume either way, and report the result in Section V-B.

Training. The deployed checkpoint uses AdamW with a head learning rate of 5×10^{-4} , backbone 10^{-6} after

staged unfreezing of the last four blocks at epoch 10, weight decay 10^{-2} , batch size 16, cosine annealing, 25 epochs with early stopping on validation macro precision, bf16 mixed precision, and six-view test-time augmentation. Every ablation arm in Table II instead shares one common recipe – 15 epochs, batch 32, cosine annealing, identical augmentation, selection on validation mAP – so that arms differ only in backbone and adaptation.

Metrics. We report AUC-ROC and AUPRC, which are threshold-free, alongside sensitivity, specificity, PPV, FNR and FPR at explicit operating points. Confidence intervals are 95% percentile bootstrap over 1,000 re-samples of the test images. Calibration is measured as expected calibration error over 15 equal-width bins.

Reporting standards. The revision follows CLAIM 2024 [17] and TRIPOD+AI [18]: one held-out split used once, per-outcome discrimination *and* calibration, interval estimates on headline figures, and an explicit intended population. Against the six FUTURE-AI principles [19], this study evidences traceability, explainability (Section V-F, including where it fails) and robustness under quantisation (Section V-E), and evidences neither fairness nor universality, which need the demographically annotated local data we do not have.

V. RESULTS

A. Per-Disease Performance

Table I gives the full per-disease breakdown. Macro AUC is 0.872 (95% CI 0.846–0.896) and macro AUPRC is 0.357 (95% CI 0.345–0.406). The aggregate hides a wide spread that matters clinically: myopia reaches an AUC of 0.996, media haze 0.974, age-related macular degeneration 0.972 and diabetic retinopathy 0.967 with an AUPRC of 0.873, while optic disc pallor (0.663), macular hole (0.674) and asteroid hyalosis (0.687) are close to uninformative. Fig. 2(b) shows the pattern: performance tracks the number of positive examples, and every class below roughly 20 test positives has a confidence interval too wide to support a clinical claim.

B. Backbone and Adaptation Ablation

Table II isolates the components. Retinal pretraining carries most of the signal: freezing RETFound and training only the head reaches AUC 0.877 while updating 0.8% of the parameters, already ahead of every ImageNet baseline. Low-rank adaptation then adds a real but modest increment, taking AUC to 0.882 and AUPRC from 0.360 to 0.392 for 1.6 M extra trainable parameters, so LoRA is worth its cost. The comparison that qualifies the story is ViT-B/16: fine-tuned end-to-end on ImageNet

initialisation it reaches a lower AUC of 0.871 but a substantially higher AUPRC of 0.472, the best of any arm. Because AUPRC is the metric that reflects performance under class imbalance, the fair conclusion is not that the retinal foundation model dominates. It gives the best ranking while training 1.3% of its parameters, which is what makes single-GPU institutional training feasible, but a fully fine-tuned ImageNet transformer retrieves rare positives better. We report this because the previous draft implied a larger and less qualified benefit than the evidence supports.

The same table settles the two preprocessing questions empirically. Replacing ImageNet normalisation with RFMiD’s own channel statistics costs 0.054 AUC and 0.107 AUPRC, confirming that matching the pre-training distribution matters more than matching the fine-tuning corpus. Adding CLAHE costs 0.016 AUC and 0.051 AUPRC. The second result supports the reviewer’s concern directly: contrast equalisation is not free on fundus images, and on this corpus it removes more diagnostic signal than it recovers, which is a reason to leave it out rather than an oversight. The post-hoc rows complete the picture. Quantisation is discussed in Section V-E; the knowledge graph moves macro AUC by -0.0002 because its reasoning rules alter only 12.7% of images and a single class, for the reason given in Section V-G.

C. What the Threshold Policy Costs

Table III answers the reviewer question about the other side of the trade-off. The deployed precision-floor policy is not, as previously described, a false-positive reduction: it is a recall-first policy. Relative to a uniform $\tau=0.5$ it reduces false negatives from 199 to 81 across the test split, a 59% reduction, but raises false positives from 484 to 3,858 and drops macro PPV from 0.383 to 0.114, taking the alert rate from 1.5 to 6.9 findings per image. It also leaves five classes permanently silent. A policy that instead fixes a sensitivity target of 0.90 on validation dominates it on the test split: macro sensitivity 0.811 against 0.591, macro PPV 0.189 against 0.114, fewer false negatives, and no silent classes.

D. Calibration

The behaviour above has a single cause. The model’s probabilities are severely over-confident: mean predicted probability is 0.251 against an empirical positive rate of 0.042, the minimum probability over the entire test split is 0.064, and expected calibration error is 0.209. Because nothing is ever confidently negative, the precision-floor search drives thresholds to the bottom of the grid for

TABLE I

PER-DISEASE PERFORMANCE ON THE HELD-OUT OFFICIAL RFMiD TEST SPLIT ($n=640$ IMAGES, 24 RETAINED CLASSES). AUC AND AUPRC ARE THRESHOLD-FREE; THE OPERATING-POINT COLUMNS USE THE PER-CLASS THRESHOLDS τ_c FITTED ON THE VALIDATION SPLIT UNDER THE DEPLOYED PRECISION-FLOOR POLICY. BRACKETS GIVE 95% PERCENTILE BOOTSTRAP CONFIDENCE INTERVALS OVER 1,000 IMAGE RESAMPLES. FNR IS THE FALSE-NEGATIVE RATE AND THE FALSE-POSITIVE RATE IS ITS COMPLEMENT ON THE NEGATIVE SIDE, $FPR = 1 - \text{Spec.}$; A DASH MARKS A QUANTITY THAT IS UNDEFINED BECAUSE THE CLASS NEVER FIRES.

Class	n_+	Prev. (%)	AUC	95% CI	AUPRC	τ_c	Sens.	Spec.	PPV	FNR
DR	124	19.4	0.967	[0.95, 0.98]	0.873	0.05	1.000	0.000	0.194	0.000
ARMMD	31	4.8	0.972	[0.95, 0.99]	0.695	0.17	1.000	0.575	0.107	0.000
MH	104	16.2	0.974	[0.96, 0.98]	0.875	0.05	1.000	0.000	0.163	0.000
DN	46	7.2	0.792	[0.73, 0.85]	0.203	0.27	0.957	0.316	0.098	0.043
MYA	32	5.0	0.996	[0.99, 1.00]	0.931	0.13	1.000	0.528	0.100	0.000
BRVO	23	3.6	0.919	[0.85, 0.97]	0.649	0.29	0.826	0.793	0.129	0.174
TSLN	53	8.3	0.940	[0.91, 0.97]	0.638	0.05	1.000	0.000	0.083	0.000
ERM	5	0.8	0.787	[0.59, 0.95]	0.049	0.37	0.200	0.986	0.100	0.800
LS	15	2.3	0.868	[0.74, 0.96]	0.189	0.31	0.800	0.811	0.092	0.200
MS	7	1.1	0.903	[0.75, 0.98]	0.147	0.37	0.714	0.942	0.119	0.286
CSR	13	2.0	0.907	[0.81, 0.97]	0.314	0.37	0.615	0.931	0.157	0.385
ODC	91	14.2	0.783	[0.73, 0.84]	0.530	0.05	1.000	0.000	0.142	0.000
CRVO	9	1.4	0.961	[0.92, 1.00]	0.578	0.27	0.778	0.919	0.121	0.222
AH	5	0.8	0.687	[0.48, 0.90]	0.029	0.95	0.000	1.000	–	1.000
ODP	24	3.8	0.663	[0.59, 0.74]	0.054	0.95	0.000	1.000	–	1.000
ODE	17	2.7	0.922	[0.82, 1.00]	0.662	0.23	0.824	0.761	0.086	0.176
AION	4	0.6	0.702	[0.38, 0.95]	0.093	0.35	0.250	0.951	0.031	0.750
PT	4	0.6	0.930	[0.86, 0.98]	0.067	0.95	0.000	1.000	–	1.000
RT	5	0.8	0.831	[0.71, 0.96]	0.039	0.35	0.200	0.964	0.042	0.800
RS	14	2.2	0.981	[0.96, 1.00]	0.619	0.19	1.000	0.859	0.137	0.000
CRS	11	1.7	0.907	[0.77, 0.98]	0.182	0.57	0.273	0.976	0.167	0.727
EDN	4	0.6	0.914	[0.85, 0.97]	0.050	0.95	0.000	1.000	–	1.000
RPEC	4	0.6	0.941	[0.86, 0.98]	0.081	0.41	0.750	0.959	0.103	0.250
MHL	3	0.5	0.674	[0.33, 0.95]	0.016	0.95	0.000	1.000	–	1.000
Macro	648	–	0.872	[0.85, 0.90]	0.357	–	0.591	0.720	0.114	0.409

RFMiD label abbreviations: DR, diabetic retinopathy; ARMMD, age-related macular degeneration; MH, media haze; DN, drusen; MYA, myopia; BRVO, branch retinal vein occlusion; TSLN, tessellation; ERM, epiretinal membrane; LS, laser scar; MS, macular scar; CSR, central serous retinopathy; ODC, optic disc cupping; CRVO, central retinal vein occlusion; AH, asteroid hyalosis; ODP, optic disc pallor; ODE, optic disc edema; AION, anterior ischaemic optic neuropathy; PT, parafoveal telangiectasia; RT, retinal traction; RS, retinitis; CRS, chorioretinitis; EDN, exudation; RPEC, retinal pigment epithelium changes; MHL, macular hole.

high-prevalence classes, where specificity collapses to zero, and to the fallback for rare ones, where sensitivity collapses to zero. Per-class Platt scaling fitted on the validation split reduces test ECE to 0.003 (macro ECE 0.210 to 0.013) and, being monotone, leaves AUC unchanged at 0.872. Fig. 2(a) shows the before and after. Applied together with the threshold policy, recalibration cuts the false-positive count from 3,986 to 2,356 and raises macro PPV from 0.189 to 0.216. We attribute the miscalibration to the asymmetric loss: probability clipping at 0.05 removes the gradient that would push confident negatives towards zero, and label smoothing bounds the targets away from the extremes.

E. Effect of 8-Bit Quantisation

Table IV compares full and 8-bit precision directly on the same 640 test images with identical CPU threading, replacing the previous unsupported claim of negligible loss. The claim holds for ranking and fails for retrieval: macro AUC is unchanged at 0.873 against 0.872, but macro AUPRC falls from 0.357 to 0.346, a 3.1% relative loss concentrated in the rare classes

whose positive examples are already scarce. At the deployed operating point the quantised model is slightly more permissive, catching eight more true positives at the cost of 218 additional false alarms. Two practical points follow. First, a screening system should be re-thresholded after quantisation rather than inheriting the full-precision thresholds, because quantisation shifts the operating point even when it preserves the ranking. Second, dynamic quantisation is a storage optimisation, not a speed one for this architecture: it cuts the exported model by 74.7% but reduces CPU latency by only 4%, because PyTorch dynamic quantisation compresses linear weights while the attention matrix products that dominate a ViT-L forward pass stay in floating point. The latency benefit in deployment comes instead from the distilled MobileNetV3 student, which runs in 102 ms per image at 5.4 MB and agrees with its own FP32 export on 99.6% of binary decisions.

F. Faithfulness of the Explanations

Table V reports the deletion/insertion test [13] on 60 confidently detected true positives, against a random-

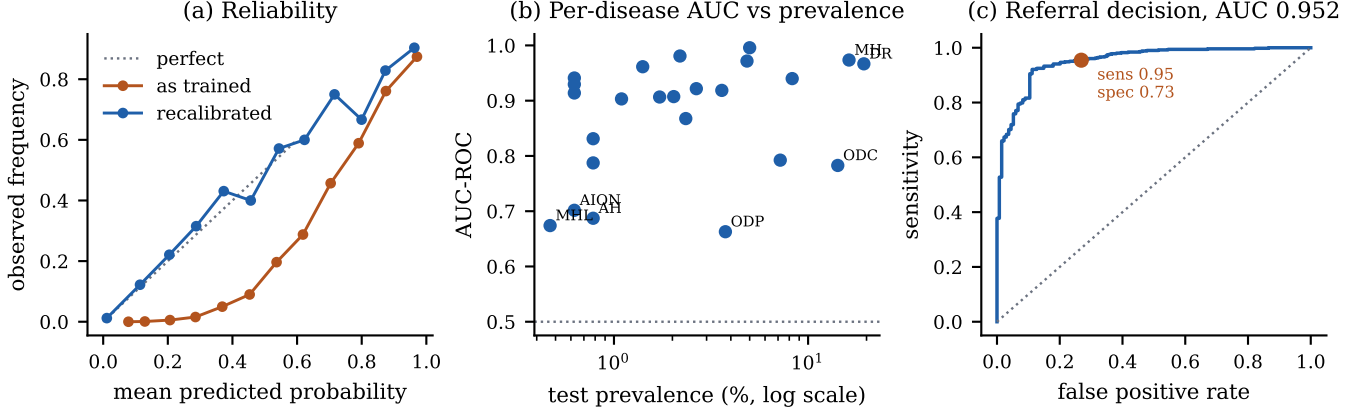


Fig. 2. (a) Reliability on the test split before and after per-class Platt scaling; the as-trained curve sits far below the diagonal, which is over-confidence. (b) Per-disease AUC against test prevalence, log scale; the weakest classes are the rarest. (c) ROC for the referral decision, with the validation-fitted high-sensitivity operating point marked.

TABLE II

COMPONENT ABLATION ON THE RFMiD TEST SPLIT. THE ADAPTATION ARMS SHARE THE DATA, THE ASYMMETRIC LOSS, A 15-EPOCH SCHEDULE, SELECTION ON VALIDATION MAP AND THE SAME THRESHOLD OPTIMISATION, SO DIFFERENCES ARE ATTRIBUTABLE TO THE COMPONENT NAMED. POST-HOC ROWS ARE APPLIED TO THE DEPLOYED CHECKPOINT, SO THEY ARE EXACT RATHER THAN RE-TRAINED. THE GRAPH ALTERS 12.7% OF IMAGES AND ONLY ONE CLASS (ODE), BECAUSE THE PARTNER CONDITIONS ITS CO-OCCURRENCE RULES ENCODE ARE MOSTLY AMONG THE 21 CLASSES DROPPED FOR HAVING TOO FEW TRAINING POSITIVES. [†]THE DEPLOYED CHECKPOINT ADDITIONALLY USES 25 EPOCHS, STAGED BACKBONE UNFREEZING AND SIX-VIEW TEST-TIME AUGMENTATION.

Arm or component	Params (M)	Train. (M)	% train.	AUC	AUPRC
<i>Backbone and adaptation (shared 15-epoch recipe)</i>					
ResNet-50 (ImageNet, full FT)	23.6	23.56	100.0	0.867	0.336
EfficientNet-B3 (ImageNet, FT)	10.7	10.73	100.0	0.819	0.377
ViT-B/16 (ImageNet, full FT)	85.8	85.82	100.0	0.871	0.472
RETFound ViT-L, head only	305.7	2.44	0.8	0.877	0.360
RETFound ViT-L + LoRA $r=16$	307.3	4.01	1.3	0.882	0.392
<i>Input preprocessing (RETFound + LoRA arm)</i>					
RFMiD-specific normalisation	—	—	—	0.827	0.285
ImageNet norm. + CLAHE	—	—	—	0.866	0.341
<i>Post-hoc components (deployed checkpoint)</i>					
+ knowledge-graph reasoning	—	—	—	0.871	0.357
+ INT8 dynamic quantisation	—	—	—	0.873	0.346
Deployed checkpoint [†]	305.7	2.44	0.8	0.872	0.357

saliency control on the same images. Both attribution methods pass the deletion test: removing the regions they highlight degrades the target probability significantly faster than removing random regions (Grad-CAM $p=0.029$, Integrated Gradients $p=0.008$, one-sided Wilcoxon). Neither passes the insertion test, where both are statistically indistinguishable from random. The maps therefore identify regions the model uses, but they do not identify a region set sufficient to reproduce the

TABLE III

WHAT THE THRESHOLD POLICY COSTS ON EACH SIDE OF THE TRADE-OFF, ON THE RFMiD TEST SPLIT. COUNTS ARE SUMMED OVER ALL 24 CLASSES AND 640 IMAGES. “SILENT” COUNTS CLASSES WHOSE THRESHOLD IS SO HIGH THAT THE CLASS NEVER FIRES, SO ITS SENSITIVITY IS IDENTICALLY ZERO.

Policy	Sens.	Spec.	PPV	FN	FP	Silent
<i>As trained</i>						
Uniform $\tau=0.5$	0.420	0.965	0.383	199	484	6
Precision floor (deployed)	0.591	0.720	0.114	81	3858	5
Sensitivity target ≥ 0.90	0.811	0.729	0.189	75	3986	0
<i>After per-class Platt recalibration</i>						
Uniform $\tau=0.5$	0.244	0.994	0.733	348	84	13
Precision floor (deployed)	0.347	0.943	0.439	194	764	11
Sensitivity target ≥ 0.90	0.657	0.835	0.216	92	2356	1

TABLE IV

EFFECT OF 8-BIT DYNAMIC QUANTISATION ON THE REFERENCE MODEL, ON THE SAME 640 TEST IMAGES WITH IDENTICAL CPU THREADING, SO ACCURACY, SIZE AND LATENCY ARE DIRECTLY COMPARABLE. THE DEPLOYED ON-DEVICE STUDENT COMPRESSES FURTHER, FROM 20.8 MB TO 5.4 MB ONNX (−73.8%), AGREEING WITH ITS FP32 EXPORT ON 99.6% OF BINARY DECISIONS AT $\tau=0.5$ AND RUNNING AT 102 MS MEDIAN PER IMAGE.

Precision	MB	AUC	AUPRC	Sens.	PPV	ms/img
FP32	1224	0.872	0.357	0.591	0.114	434
INT8 dynamic	310	0.873	0.346	0.610	0.106	418
Change	−74.7%	+0.002	−0.011	+0.018	−0.008	−4%

prediction, and the flat insertion curves are a direct consequence of the calibration problem: predicted probabilities never fall below 0.064, so revealing evidence has little room to move them. We accordingly present the maps as attention hints requiring clinician verification, not as standalone evidence. The narrative layer is audited

TABLE V

FAITHFULNESS OF THE EXPLAINABILITY LAYER ON 60 CONFIDENTLY DETECTED TRUE POSITIVES. DELETION AUC IS BETTER WHEN LOWER, INSERTION AUC WHEN HIGHER; p -VALUES ARE ONE-SIDED WILCOXON SIGNED-RANK TESTS AGAINST THE RANDOM-SALIENCY CONTROL ON THE IDENTICAL IMAGES. NARRATIVE LAYER ($n=24$ REPORTS FROM THE DISTILLED NARRATOR THAT ALWAYS GENERATES AND NEVER OMITTS A DETECTED FINDING): PROBABILITY INFIDELITY 0% (95% CI 0%–14%), FINDING OMISSION 0%, UNSUPPORTED SEVERITY ESCALATION 21% (95% CI 9%–40%).

Attribution	Del. AUC	vs. rand.	Ins. AUC	vs. rand.
Grad-CAM	0.913	$p = 0.029$	0.926	$p = 0.996$
Integrated Grad.	0.861	$p = 0.008$	0.943	$p = 0.590$
Random (control)	0.946	–	0.943	–

separately for probability infidelity, omission of detected findings, and unsupported escalation of urgency.

G. Referral Triage and What the Knowledge Graph Adds

Per-disease multi-label metrics understate the system, because the output a primary-care operator acts on is a single referral decision. Scored against the RFMiD Disease_Risk flag, the maximum class probability reaches an AUC of 0.952 (95% CI 0.932–0.968) and an AUPRC of 0.986. At the validation-fitted high-sensitivity operating point, test sensitivity is 0.955 and specificity 0.731, missing 23 of 506 positive images.

The knowledge graph’s measurable contribution is here, not in classification accuracy. Its co-occurrence rules almost never fire on the retained label set, because most of the partner conditions they encode are among the 21 classes dropped for having too few training examples; claiming a macro-F1 improvement from the graph, as the previous draft did, is not supportable. Its referral-priority rules, however, do stratify risk monotonically once the thresholds are non-degenerate: under the sensitivity-targeted policy the graph routes 84.5% of images to URGENT (86.9% of which carry pathology), 10.8% to ROUTINE (47.8%), and 4.7% to FOLLOW-UP (10.0%). Under the shipped thresholds the same rules classify 100% of images as URGENT and carry no information at all, which is a further consequence of the calibration failure.

H. From Benchmark to Clinic

Table VI is the reason this paper does not claim clinical readiness. RFMiD is an enriched research corpus in which 79% of test images carry pathology. Holding the measured sensitivity and specificity fixed and varying only prevalence, PPV falls from 0.931 on the

TABLE VI

WHY A BENCHMARK RESULT IS NOT A CLINICAL RESULT. THE REFERRAL OPERATING POINT IS HELD FIXED AT THE MEASURED TEST SENSITIVITY 0.955 AND SPECIFICITY 0.731; ONLY THE PREVALENCE OF THE SCREENED POPULATION CHANGES. RFMiD IS AN ENRICHED RESEARCH CORPUS, SO ITS POSITIVE RATE IS FAR ABOVE ANY PRIMARY-CARE CASELOAD.

Population	Prev. (%)	PPV	NPV	Refer (%)
RFMiD test (as measured)	79	0.931	0.810	81
Projected at 30% prev.	30	0.604	0.974	47
Projected at 20% prev.	20	0.470	0.985	41
Projected at 10% prev.	10	0.283	0.993	34
Projected at 5% prev.	5	0.158	0.997	30

benchmark to 0.470 at 20% prevalence and 0.158 at 5%, while roughly 30% of all screened patients would still be referred. A screening programme sized on the benchmark PPV would therefore under-provision its referral pathway by a factor of three or more. Negative predictive value moves the other way, exceeding 0.99 at low prevalence, which is the property a rule-out triage tool most needs.

VI. DISCUSSION AND LIMITATIONS

We state concretely that benchmark results on international corpora cannot be interpreted as demonstration of clinical efficacy in Uganda. RFMiD reflects predominantly South Asian demographics imaged on table-top hospital fundus cameras. Zero Ugandan patient images were evaluated; hence, findings cannot guarantee robustness across local retinal pigmentation, systemic comorbidities (e.g., HIV microangiopathy), or hand-held smartphone-adapted cameras used in district health centres. Our prevalence transfer model (Section V-H) quantifies the referral burden shift; optical distribution shifts remain unmeasured.

Three methodological constraints bound our claims. First, rare categories have limited statistical power: 21 of 45 RFMiD labels were dropped for having fewer than 10 training positives, and conditions with fewer than 20 evaluation positives exhibit wide bootstrap intervals. Second, the clinical knowledge graph (Section III-D) is curated from published evidence but lacks independent ophthalmologist panel review. Third, ablation arms share a standardized 15-epoch schedule, bounding relative contributions rather than ceiling performance.

Clinical translation requires prospective multi-centre trials in Ugandan district facilities against local ophthalmologist reference standards. Uganda National Drug Authority clearance, Data Protection and Privacy Act (2019) compliance, ISO 13485 quality systems, and

Good Machine Learning Practice alignment remain prerequisites. An interactive demonstration is hosted at <https://landwind22-retinal-screening.hf.space> as a workflow prototype, not a cleared device.

VII. CONCLUSION

We evaluated OptiscanAI on the held-out RFMiD test split and report what the evidence supports. Discrimination is strong (macro AUC 0.872; referral triage AUC 0.952), yet ablation reveals that retinal foundation pretraining primarily confers parameter efficiency, while a fully fine-tuned ImageNet transformer attains higher rare-class retrieval (AUPRC 0.472 vs 0.392). The shipped threshold policy is recall-prioritising and silences five classes due to severe probability overconfidence (ECE 0.209); rank-preserving Platt scaling resolves this failure mode (ECE 0.003). Attribution maps pass deletion but fail insertion fidelity testing. Crucially, transferring triage operating points to primary-care prevalence cuts PPV by over half. The work positions itself deliberately as an initial step toward responsible, context-appropriate retinal screening, with prospective clinical validation on local cohorts, multi-stakeholder governance, and regulatory clearance as non-negotiable preconditions for clinical deployment.

ACKNOWLEDGMENTS

The authors express gratitude to the faculty and research leadership at Makerere University's College of Computing and Information Sciences for technical counsel throughout this investigation. We acknowledge the curators of the public RFMiD repository for releasing their annotated dataset to the research community. Furthermore, we sincerely thank the anonymous conference reviewers; their insightful critiques concerning class-specific metrics, modular attribution, and demographic framing directly informed this revised camera-ready manuscript.

REFERENCES

- [1] World Health Organization, "World report on vision," World Health Organization, Geneva, Switzerland, Tech. Rep., 2019.
- [2] Z. L. Teo, Y.-C. Tham, M. Yu, *et al.*, "Global prevalence of diabetic retinopathy and projection of burden through 2045: Systematic review and meta-analysis," *Ophthalmology*, vol. 128, no. 11, pp. 1580–1591, 2021.
- [3] V. Gulshan, L. Peng, M. Coram, *et al.*, "Development and validation of a deep learning algorithm for detection of diabetic retinopathy in retinal fundus photographs," *JAMA*, vol. 316, no. 22, pp. 2402–2410, 2016.
- [4] E. Beede, E. Baylor, F. Hersch, *et al.*, "A human-centered evaluation of a deep learning system deployed in clinics for the detection of diabetic retinopathy," in *Proc. CHI Conf. Human Factors in Computing Systems*, 2020, pp. 1–12.
- [5] J. Wiens, S. Saria, M. Sendak, *et al.*, "Do no harm: A roadmap for responsible machine learning for health care," *Nature Medicine*, vol. 25, no. 9, pp. 1337–1340, 2019.
- [6] S. Pachade, P. Porwal, D. Thulkar, *et al.*, "Retinal fundus multi-disease image dataset (RFMiD): A dataset for multi-disease detection research," *Data*, vol. 6, no. 2, p. 14, 2021.
- [7] A. Dosovitskiy, L. Beyer, A. Kolesnikov, *et al.*, "An image is worth 16×16 words: Transformers for image recognition at scale," in *Proc. Int. Conf. Learning Representations (ICLR)*, 2021.
- [8] Y. Zhou, M. A. Chia, S. K. Wagner, *et al.*, "A foundation model for generalizable disease detection from retinal images," *Nature*, vol. 622, pp. 156–163, 2023.
- [9] E. J. Hu, Y. Shen, P. Wallis, *et al.*, "LoRA: Low-rank adaptation of large language models," in *Proc. Int. Conf. Learning Representations (ICLR)*, 2022.
- [10] R. R. Selvaraju, M. Cogswell, A. Das, *et al.*, "Grad-CAM: Visual explanations from deep networks via gradient-based localization," in *Proc. IEEE Int. Conf. Computer Vision (ICCV)*, 2017, pp. 618–626.
- [11] S. M. Lundberg and S.-I. Lee, "A unified approach to interpreting model predictions," in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 30, 2017, pp. 4765–4774.
- [12] M. Sundararajan, A. Taly, and Q. Yan, "Axiomatic attribution for deep networks," in *Proc. Int. Conf. Machine Learning (ICML)*, 2017, pp. 3319–3328.
- [13] V. Petsiuk, A. Das, and K. Saenko, "RISE: Randomized input sampling for explanation of black-box models," in *Proc. British Machine Vision Conf. (BMVC)*, 2018.
- [14] T.-Y. Lin, P. Goyal, R. Girshick, *et al.*, "Focal loss for dense object detection," in *Proc. IEEE Int. Conf. Computer Vision (ICCV)*, 2017, pp. 2980–2988.
- [15] T. Ridnik, E. Ben-Baruch, N. Zamir, *et al.*, "Asymmetric loss for multi-label classification," in *Proc. IEEE Int. Conf. Computer Vision (ICCV)*, 2021, pp. 82–91.
- [16] C. Guo, G. Pleiss, Y. Sun, *et al.*, "On calibration of modern neural networks," in *Proc. Int. Conf. Machine Learning (ICML)*, 2017, pp. 1321–1330.
- [17] A. S. Tejani, M. E. Klontzas, A. A. Gatti, *et al.*, "Checklist for Artificial Intelligence in Medical Imaging (CLAIM): 2024 update," *Radiology: Artificial Intelligence*, vol. 6, no. 4, e240300, 2024.
- [18] G. S. Collins, K. G. M. Moons, P. Dhiman, *et al.*, "TRIPOD+AI statement: Updated guidance for reporting clinical prediction models that use regression or machine learning methods," *BMJ*, vol. 385, e078378, 2024.
- [19] K. Lekadir, A. Frangi, A. R. Porras, *et al.*, "FUTURE-AI: International consensus guideline for trustworthy and deployable artificial intelligence in healthcare," *BMJ*, vol. 388, e081554, 2025.